LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Uploaded Diabetes Dataset

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Objective	Dataset	Model
Predict a binary diabetes-related outcome: 0 = No Diabetes, 1 = Diabetes.	403 records and 19 columns. 13 records with missing glyhb were removed before creating the target.	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: chol, stab.glu, hdl, ratio, location, age, gender, height, weight, frame, bp.1s, bp.1d, bp.2s, bp.2d, waist, hip, and time.ppn. **Target:** Outcome is created from glyhb using glyhb ≥ 6.5 → 1 and glyhb < 6.5 → 0. **Workflow:** load CSV → inspect data → remove records with missing glyhb → create Outcome → remove id/glyhb from predictors → one-hot encode categorical variables → train-test split → median imputation → scaling → balanced Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.846	Overall correct predictions
Precision	0.524	Reliability of positive predictions
Recall	0.846	Actual positive cases detected
F1-score	0.647	Balance of precision and recall
ROC-AUC	0.925	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 55	FP = 10
Actual 1	FN = 2	TP = 11

Key Findings

The model achieved an **ROC-AUC of 0.925** and a **recall of 0.846** on the fixed test split (78 records). The strongest positive Logistic Regression coefficient was **stab.glu (1.973)**, followed by waist (0.739), age (0.680), bp.1s (0.489), and chol (0.426). The confusion matrix contains 55 true negatives, 10 false positives, 2 false negatives, and 11 true positives.

Limitation and Conclusion

The target in this adapted project is derived from glyhb rather than an original Outcome column, using the 6.5 threshold implemented in the notebook. Therefore, the results should be interpreted as an educational classification exercise. The model demonstrates the complete Logistic Regression workflow, including preprocessing, evaluation, and coefficient interpretation. It is not intended for medical diagnosis.

Submission checklist: Submit the executed .ipynb file and this PDF report. Keep the metric outputs, confusion matrix, ROC curve, and coefficient graph visible in the notebook.